

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

88
100

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

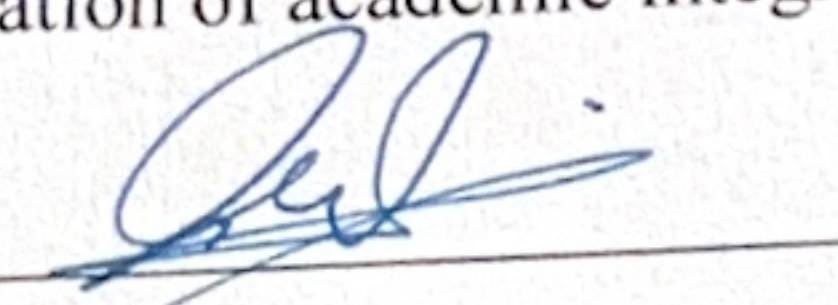
Duration: 60 Minutes

Total Marks: 25

Code of Conduct

I R. Ram Nitesh (1925113027) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Assessment Tasks

Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services..
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

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RegNo : 192511302

Course : Cloud Computing and Big data analytics

Code : CSA1521

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Assessment Tool - I

Short Answer test

Describe the evolution from standalone computing to cloud based systems. Summarize key technological transitions that enable cloud computing.

i) Standalone Computing (1960s)

- * Each computer worked independently with local storage and software

ii) Client-Server Architecture (1970s)

- * Clients accessed centralized servers for data and applications over a network

iii) Distributed Computing (1980s)

- * Multiple computers could run and work together

iv) Grid Computing (1990s)

- * Computing resources were shared and provided as utility services.

v) Cloud Computing (2000s)

- * On-demand access to computing resources over the Internet with scalability and pay-as-you-go pricing

Key Technological Transitions

- i) Local Processing only
- ii) Centralized data and application management
- iii) Global Connectivity and remote access.
- iv) Efficient use of hardware through virtual machines.
- v) Cost effective Computing Services

2.

Interpret the working of hypervisors in virtualization
Differentiate between Type 1 and Type 2 hypervisors with examples.

⇒

Working of a hypervisor

* A hypervisor is software that creates and manages virtual machines. It sits between the physical hardware and the virtual machines, allocating CPU, memory, storage and network resources to each VM. This allows multiple operating systems to run simultaneously on a physical computer.

Type 1 Hypervisor
(Bare metal)

Runs directly on physical hardware

Higher performance and security

Mainly used in data centres and cloud environments

Eg: VMware, ESXi, Xen

Type 2 Hypervisor
(Hosted)

Runs on top of a host operating system

Lower performance due to host OS

Mainly used for PC, testing, and development

Eg: Oracle, VirtualBox

Describe the role of APIs in cloud service integration.
Explain how APIs support interoperability between services.

⇒

Role of API

* An API (Application Programming Interface) is a set of rules that allows different cloud applications and services to communicate and exchange data. APIs enable developers to integrate multiple cloud services without needing to know their internal implementation.

Working diagram

cloud service A

↓ API Request

API

↓ API response

cloud service B

How APIs Support Interoperability

- * Enable Communication between different cloud applications
- * Allow data sharing using standard protocols such as REST and SOAP
- * Connect services from different cloud providers.
- * Automate tasks to improve system integration
- * Reduce development time by reusing existing services

Example :

* An e-commerce application uses a payment gateway API to process online payments and a shipping API to track deliveries. These APIs allow different services to work together seamlessly.

4.

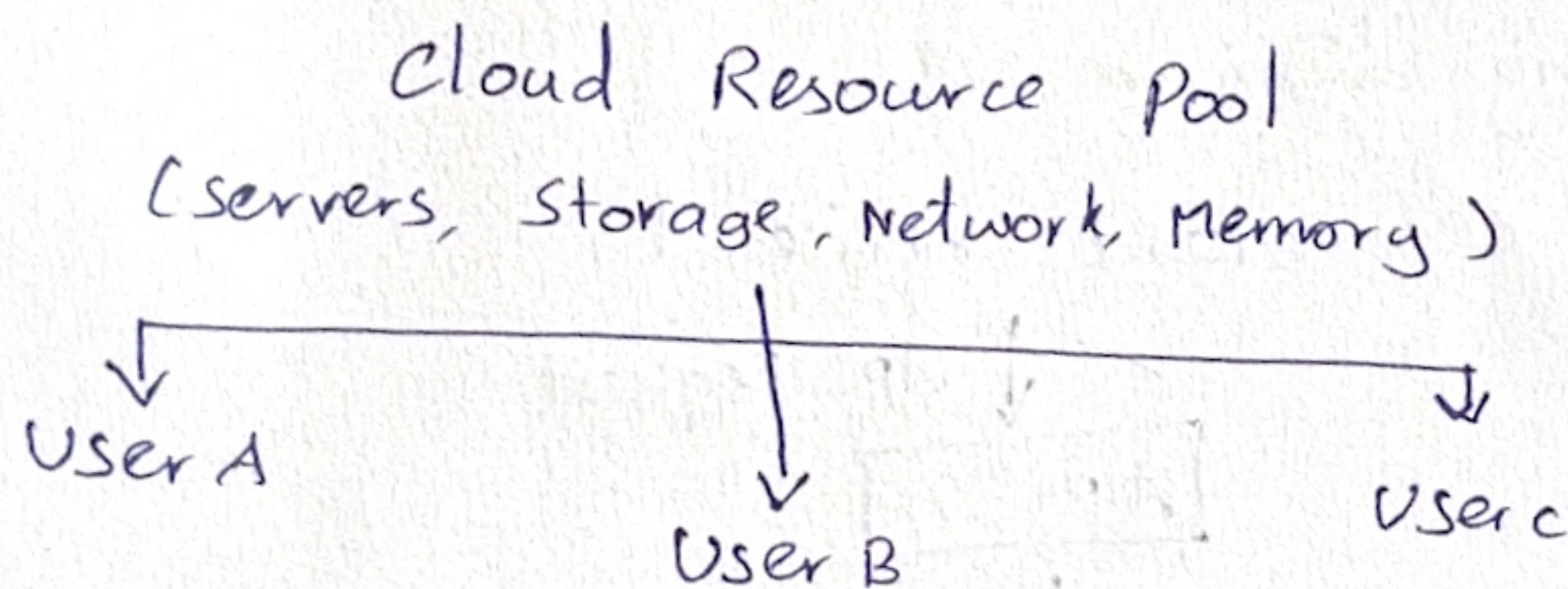
Explain the concept of resource pooling in cloud computing. illustrate how it benefits multiple users.

⇒

Resource Pooling

* It is a cloud computing feature in which the cloud provider combines computing resources such as servers, storage, memory and network into a shared pool. These resources are dynamically assigned & reassigned to multiple users based on their demand.

Working diagram



Benefits to Multiple Users

- * Resources are shared efficiently among many users.
- * Reduces infrastructure and maintenance costs.
- * Provides high availability and better resource utilization.
- * Resources are allocated dynamically based on user demand.
- * Each user's data remains secure and isolated, even though resources are shared.

Example:

* A cloud provider like AWS or Microsoft Azure uses the same physical servers to host applications for multiple companies. Each company gets the resources it needs while remaining isolated from other users.

Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workload
⇒

* It is ability of cloud computing to automatic-ally increase or decrease computing resources according to workload demand.

Working

- * Resources are mentioned continuously
- * when demand increases, additional servers allotted
- * when demand decreases, unused resources are released automatically

Benefits

- * Reduces operational cost
- * Prevents resource wastage
- * Improves performance
- * Ensures high availability

Example

During an online shopping sale cloud resources automatically increase to support millions of users. At end it removed

Dynamic workload

Video streaming

online exam

E-commerce